Self Without a Transcript: Semantic Recurrence Across Text and Latent States

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Abstract

What can a frozen language model recover after a description is encoded into eight hidden-state slots? We compare text relay, latent recurrence and a language roundtrip inserted after the same latent update. In Mistral-7B-Instruct-v0.3, allowing generation to reach native EOS still produced one roundtripped state across two opposed initial descriptions. Across six further development contrasts, Mistral and Gemma-2-2B-IT each identified 24/24 literal descriptions. Each checkpoint produced 24 distinct updated tensors but one greedy readout string. In a lexical positive control with matched sequence lengths, supplying only eight middle-layer donor slots to a neutral recipient gave 6/12 correct preferences on Mistral and 7/12 on Gemma. Supplying downstream donor positions that had already processed the question reproduced all 12 native preference signs on both checkpoints. These interventions measure access to explicit content at a particular boundary. Semantic qualification of the recurrent state and transfer to new tasks remain open.

1 Problem formulation

A description such as "I prioritize keeping decisions revisable" may be reinterpreted as a preference for preserving correction opportunities. The transition may add a useful distinction, remain a paraphrase, or introduce an unsupported claim. Our object is the change in meaning, not the event of uttering the earlier description. We exclude a direct transcript and instructions that infer character from the act of speaking; this does not erase past updates' causal influence on the current state.

We ask whether recurrence yields defensible distinctions, whether a language roundtrip changes their formation or use, and whether changes improve held-out problem solving or appropriate switching under reversed conditions. Convergence is not intrinsically failure, and novelty is not intrinsically success. "Self" denotes an attribution frame rather than a claim about subjective experience.

2 Related work

Coconut trains continuous-state reasoning through hidden-state feedback [5]. Recurrent-depth models study repeated internal computation [4]; STARS motivates distinguishing useful stable dynamics from collapse [13]. These do not establish the validity of arbitrary untrained residual feedback.

Persona vectors and the Assistant Axis connect persona-related behavior with activation structure and interventions [1, 9]. Natural Language Autoencoders provide a learned activation/text

roundtrip [3], while latent chain-of-thought analysis distinguishes output and representational commitment [8]. Our estimand concerns a specific codec rather than "language versus pure thought."

Semantic diversity differs from surface variation [11]. Contrasting self-correction and condition-verification findings motivate operation-specific hypotheses [6, 12]. Dadfar's report is considered with McCardle's open replication critique concerning length, mode and cycle confounds [2, 10]. SlimMoE informs the proposed Phi MoE comparison [7]. We claim neither priority nor a systematic review.

3 Three conditions

Let c_n be a fixed-schedule operation instruction, s_n a current description, and $Z_n \in \mathbb{R}^{K\times d}$ a state. With encoder E, verbalizer D, and state updater G:

$$\text{T:} \quad s_{n+1} = T_\theta(c_n, s_n), \tag{1}$$

$$\text{L:} \quad Z_{n+1} = G_\theta(c_n, Z_n), \tag{2}$$

$$\text{B:} \quad Z_{n+1} = E_\theta(D_\theta(G_\theta(c_n, Z_n))). \tag{3}$$

L and B begin from copies of $E(s_0)$ and share weights, boundaries, transport, generation policy and schedule. Their contrast is primary; T/L is a system-level comparison.

Each outer step uses a fresh context without persistent KV cache. Only current text or slots carry forward. Observation uses clones and restores RNG state; its text is not fed back except as prescribed by B. The initial description is not repeatedly reintroduced. Slots occupy explicit decoder boundaries; post-writer to pre-cut transport is explicit. An identity map is an unqualified engineering diagnostic.

The two schedules are an ordinary meaning update and a fixed sequence of decomposition, boundary examination and reintegration. Unchanged content is allowed when no justified update exists. Adaptive cycle-triggered feedback would be a separate condition.

4 Qualification and measurement

Native parity, content discrimination, contrastive-state swap effects, one-step validity, readout fidelity and positional controls precede any latent pilot. Transport or codec training may be needed; training material must remain disjoint from confirmation. One-step success does not establish long-trajectory validity.

Meaning. Blind paired judgments assess support, update type, new distinctions and event-recovery violations. Lexical and tensor distances are not semantic scores.

Activations. Selected residual, attention, MLP and supported MoE gate outputs are retained. Gate logits are not actual dispatch frequencies. The probes below test source discrimination and intervene on residual positions.

Effects. Problems arrive only after recurrence and operate on isolated copies. Near, structural and less-related transfer are separate tiers. Initial-state, one-shot instruction and non-self controls are planned. Reversals distinguish commitment from rigidity. Frozen-weight state-conditioned gains are not permanent learning.

At a shared state Z, one branch continues with $G^r(Z)$ and another with $G^r(E(D(Z)))$. The development probes below add slot order and magnitude controls. Compression matched for distance, no-update roundtrips and independent semantic review remain open.

5 Development design

The first experiment compared native EOS with an eight-token ceiling on two Mistral descriptions about revisable or prompt decisions. We subsequently restored the archived literal, encoded, latent and roundtripped states and asked which priority each donor favored. Before collecting new scores, we fixed six additional topics with two opposing preferences and two phrasings per preference. They covered filing order, walking routes, meeting notes, temperature units, shelving and directions. For each topic all donors received the same two alternatives. The question and alternatives were absent during encoding, the single plain update and greedy verbalization.

For each donor we scored both answer labels and reversed their option assignments. The plus margin is the mean, over the two orders, of the log probability of the label naming the plus option minus that of the label naming the minus option. A positive margin favors the plus option. A pair is ordered if the plus donor has a larger margin than the minus donor. Correct identification of both donors additionally requires the two margins to straddle zero in the expected direction. Ties use the stored scores at their recorded precision. Each checkpoint contributed 12 matched pairs in six topic clusters. All results are descriptive. Alternative orders and phrasings are repeated measures.

We scored literal text, E, G after one update, B after re-encoding, reversed G rows, and G vectors with the donor's per-slot magnitudes on fixed directions. Zero vectors and uninjected native placeholders served as null inputs. For each of 24 descriptions we collected D(E) and D(G), and we generated from both nulls once. Both checkpoints used one RTX 5090, eager BF16, eight slots, uncalibrated identity transport, a common 1024-token context and their native EOS ids. The cut and writer were layers 16 and 31 for Mistral and layers 13 and 25 for Gemma (zero-indexed). There was no training, prompt selection against these scores or human semantic rating.

6 Results

6.1 Native EOS and the language roundtrip

Two opposed descriptions emphasized revisable or prompt decisions. Each underwent two plain updates in T, L and B, once with native EOS and once with an eight-token ceiling. The four trials used greedy BF16 Mistral-7B-Instruct-v0.3, eight slots, a 1024-token context, a pre-layer-16 cut and a post-layer-31 writer (zero-indexed). All 18 generations per policy include main-path, observer and branch readouts. All 18 reached EOS with the full context available, and all 18 reached the explicit ceiling under the cap. With EOS, each of the four B updates verbalized the same 13-token sentence about workspace contents. The four B states matched within each policy, although the two policies gave different states. The four L states differed within each policy and matched step for step across policies.

6.2 Reading the archived state donors

For the previously studied Mistral pair, the literal revisability and speed descriptions yielded plus margins of +16.766 and -16.344 nats. The two initial slot states both gave +2.844. The first and second L updates gave distinct fingerprints but positive margins for both donors. B gave +2.813

for both donors at both steps. This particular question separated the two literal descriptions and favored revisability for every archived latent donor.

6.3 Six contrasts and matched controls

On both checkpoints all 24 literal descriptions had the correct sign. At E and G, none of the 12 matched pairs had both signs correct on either model. The B margins tied in all 12 pairs on each model. Each checkpoint had 24 distinct E tensors and 24 distinct G tensors, but just one E readout string and one G readout string per checkpoint and stage. Re-encoding D(G) produced one B tensor on each checkpoint. All 50 free generations per checkpoint ended at native EOS. Gemma's G readout requested a description to summarize, as did its zero-state readout. Mistral's G readout referred to the contents of the work areas.

Table 1: Ordered/tied pole pairs (of 12) and correct donor signs (of 24). Each pair shares its question and counterbalanced labels.

Route	Mistral pairs	Gemma pairs	Mistral signs	Gemma signs
Literal	12/0	12/0	24	24
E	6/2	5/1	12	12
G	3/0	7/1	12	11
B	0/12	0/12	12	12
G, reversed	5/2	5/0	12	13
G, magnitude only	9/0	3/2	12	12

The row-reversal and fixed-direction magnitude shams produced some positive rankings even when most donor signs were unchanged. The absolute magnitude of a state's response matters for interpreting these rankings. For ordinary G the median signed pole gap was -0.016 nats on Mistral and approximately zero on Gemma, compared with +29.680 and +20.766 for literal text. Per-topic opposite-pole and same-pole paraphrase differences, exact ties and both raw label orders are included in the supplement.

6.4 Matched lexical donors and replay geometry

A lexical positive control placed the words of one answer option at the eight designated positions, padded on the right with the neutral period token. No answer letter was included. We compared native scoring, ordinary E on the same words, and injection of residuals captured at the same positions in a prefix ending at the last slot. Native words yielded 11/12 correct donor signs on Mistral and 12/12 on Gemma. Ordinary E gave 6/12 on each. The short-prefix aligned donor gave 6/12 and 7/12. Replaying neutral slots captured from a shorter forward changed individual label log probabilities by up to 0.1903 and 0.1613 nats. Those discrepancies motivated a separate length-matched experiment.

We appended one neutral dummy token to every scoring prompt and read the logit vector immediately before it. This reproduced all earlier native label scores exactly. The eight-slot donor now used a full-length forward, with neutral tokens replacing the question and alternatives after the slots. Neutral replay then matched every native vocabulary logit in all 12 question/order combinations on both checkpoints. The opaque-tail lexical donor's slots also matched its natural full-length counterpart exactly in all 24 donor/orders per checkpoint.

Table 2: Correct donor signs and median signed pole gap in nats. The same six paired option phrases are used for each intervention.

Intervention	Mistral correct	Gemma correct	Mistral gap	Gemma gap
Native words	11/12	12/12	28.219	19.672
Eight slots	6/12	7/12	0.008	0.219
Downstream span	11/12	12/12	27.594	19.734
Both regions	11/12	12/12	28.219	19.672

We next transplanted the natural lexical donor's middle-layer residuals from the positions after the slots through the scoring position. These downstream residuals had already processed the question and alternatives. Transplanting only this region into a neutral recipient reproduced the native preference sign in all 12 cases per checkpoint, including Mistral's error on the author-shelving donor. Its full logit vectors could still differ. Transplanting the slots and downstream region together recovered the entire native scoring-position logit vector exactly in 24/24 donor/orders per checkpoint.

Supplementary plans, data and analyses.

7 Interpretation and limitations

In these development fixtures, relaxing the token ceiling changed the text and the re-encoded state while the roundtrip still mapped opposing donors to one state. The fixed question could read explicit lexical content. The standard encoded state and the matched eight-slot transplant usually failed to recover opposite preference signs. At the measured cut, the downstream-span intervention sufficed to reproduce the native choices. That span contains more positions and comes from a donor that saw the actual question. Its success does not identify a unique storage location or establish a reusable semantic representation.

The study has two frozen checkpoints, six hand-authored topics, one selected cut per model, greedy decoding, BF16 arithmetic and an uncalibrated transport. The replay experiment isolated an input-length effect under these settings. It did not identify the responsible numerical kernel. Small direction rankings should be read with the raw margins and controls. Independent evaluation of one-step meaning updates, blind judgments of new distinctions and held-out transfer remain open. No semantic gate has been approved.

8 Proposed pilot and analysis

Six content groups, two poles, two paraphrases and self/other frames yield 48 development seeds. Three arms and two schedules yield 288 proposed trajectories per model. Candidate families are Mistral-7B-Instruct-v0.3, Llama-3.2-3B-Instruct, Gemma-2-2B-IT and Phi-mini-MoE, with Phi-tiny-MoE a separately reported fallback. Family differences are not controlled architectural interventions.

Twelve steps with fixed assessment checkpoints are proposed. The present CLI scores tasks at baseline/final only. Its 20 synthetic tasks are exposed development fixtures, not a broad benchmark. Confirmation data, thresholds, sample size and preregistration are pending.

Paired L/B utility differences at a declared depth will be reported alongside support and switching. Inference must account for dependence among meaning groups and task families rather than treating each step as independent. Same-step designs report actual compute; matched-compute

analyses are secondary. Failures, censored generations and ties retain denominators. Best-depth selection on confirmation is prohibited.

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